Problem Set 6 Solutions

QTM 220

Due 1 December at 2:00pm ET on Canvas. Questions about submission requirements should be directed to your TA.

California CPS Revisited

(a) Using the California sample of 25-35 year olds, regress income on education and age (in an additive model). Report your results in a table, including the standard errors that R automatically calculates.

```
cps<- read.csv("California_CPS_degrees.csv")

mod1<- lm(income~ education + age,data=cps)

summary(mod1)
```

```
Call:
lm(formula = income ~ education + age, data = cps)

Residuals:
    Min      1Q  Median      3Q     Max
-226476  -62493  -20343   36193  752685

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -137806.3    24024.5  -5.736  1.1e-08 ***
education     16479.0      889.8  18.520  < 2e-16 ***
age            1051.6      694.4   1.514     0.13
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 101800 on 2243 degrees of freedom
Multiple R-squared:  0.1345,    Adjusted R-squared:  0.1337
F-statistic: 174.2 on 2 and 2243 DF,  p-value: < 2.2e-16
```

(b) Create a matrix of your X information for your model (education and age) and include a column of 1s to estimate an intercept. Using the following matrix functions in base R, calculate $(X'X)^{-1}X'Y$ and confirm your resulting $\hat{\beta}s$ are the same as your lm results:

Let X be your matrix in R. Use $\% * \%$ to multiply matrices, i.e. $X\% * \%X$ will yield XX

Use $t(X)$ to take the transpose of matrix X

Use solve(X) to take the inverse of matrix X

```r
df.mat<- data.frame(constant=rep(1,length(cps$income)), edu=cps$education, age=cps$age)

X<- data.matrix(df.mat)

solve(t(X)%*%X)%*%t(X)%*%(cps$income)
```

```
                [,1]
constant -137806.340
edu        16478.997
age         1051.582
```

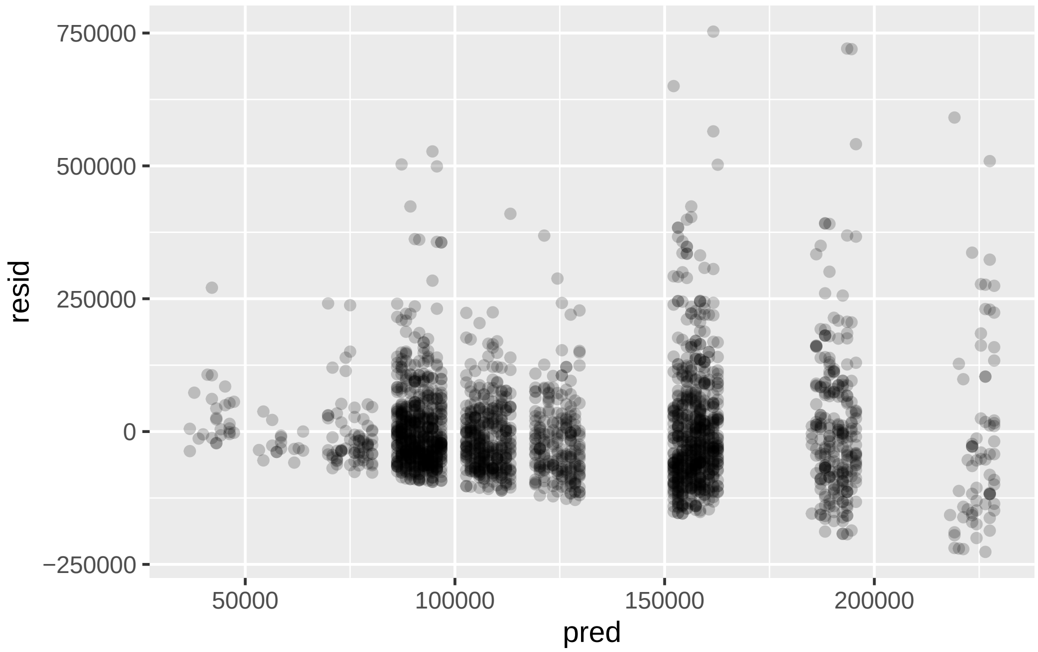
(c) Extract the predicted values and residuals from your model using predict(model name) and residuals(model name), then create a scatter plot of your predicted values and residuals. What do you see? Would you guess the errors are homoskedastic or heteroskedastic? Why?

```r
pred<- predict(mod1)
resid<- residuals(mod1)

df<- data.frame(pred=pred,resid=resid)


rplot<- ggplot(df, aes(x=pred,y=resid)) + geom_point(alpha=.2)

rplot
```



(d) install the sandwich package and get heteroskdasticity consistent standard errors for your model

```
sandwich(mod1)
```

```
              (Intercept)      education            age
(Intercept)     666941234 -15029237.613 -15481152.468
education       -15029238   1118548.385      2254.142
age             -15481152      2254.142    517429.329
```

```
sqrt(diag(sandwich(mod1)))
```

```
(Intercept)   education         age
 25825.2054   1057.6145    719.3256
```

(e) do matrix algebra to get the same thing

```
D<- diag(resid^2)

dim(D)
```

```
[1] 2246 2246
```

```
betavar<- solve(t(X)%*%X)%*%t(X)%*%D%*%X%*%solve(t(X)%*%X)

sqrt(diag(betavar))
```

```
  constant        edu        age 
25825.2054  1057.6145   719.3256 
```

(f) Bootstrap!

(g) Compare the standard errors you got from R in part (a) to the robust (sandwich) you calculated in parts (e) and (f). Why are they different? How do these compare to what you estimated from the bootstrap?

Potential outcomes review???

potential outcomes table– use conditional randomization, show that ATE among groups is the same as interactive model, show wrong model gives wrong answer, show APD=ATE

Neto and Cox

Potential outcomes table ?

Basic APD

Propensity score matching

make a balance table

Menopausal Hormone Therapy

compare RCT and observational studies

read some stuff

reflect